

Lovely Professional University



Predictive Analytics (INTM591)

Project

SUBMITTED TO:
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Project Definition:

This project aims to use predictive analytics to identify customers at risk of churning. By analyzing historical data with a CHAID decision tree model in IBM SPSS Modeler, the objective is to produce a targeted list of high-risk individuals. This allows businesses to implement proactive retention strategies and effectively lower customer attrition rates.

Outcomes/Learning:

The project provided a comprehensive learning experience in predictive analytics by utilizing the CHAID (Chi-square Automatic Interaction Detection) algorithm on a practical customer churn dataset. Through the use of IBM SPSS Modeler, I developed skills in constructing, validating, and analyzing decision tree models. Furthermore, this project demonstrated how to convert complex analytical findings into strategic business recommendations, specifically for pinpointing high-risk clients and informing data-backed retention initiatives.

Required Tool:

IBM SPSS Modeler

Working:

The project involves importing customer data, preprocessing variables, building a predictive model, and identifying high-risk churn customers. The final output is a refined list of customers with a high probability of churn for business intervention.

Summary:

Because customer churn directly impacts long-term profitability and lifetime value, this project utilizes predictive analytics to address the issue. By employing a CHAID decision tree model to analyze demographic and behavioral data, the project effectively classifies churn probability across the customer base. Using a confidence threshold of 0.65, a targeted list of high-risk individuals was generated, providing the business with a focused and actionable foundation for proactive retention strategies.

Introduction:

In today's competitive business landscape, the cost of keeping current customers is significantly lower than that of winning over new ones. Organizations can proactively manage customer churn by using predictive modeling to understand behavioral trends and intervene before a customer leaves. By successfully reducing attrition, companies can ensure more stable revenue streams and higher overall customer loyalty. This project explores the application of data mining strategies to uncover churn patterns and provide actionable insights for informed business decisions.

Data Description:

The dataset consists of customer demographic and behavioral variables that influence churn. These variables were selected based on their relevance to customer engagement and spending patterns.

- Age: Indicates customer lifecycle stage and purchasing behavior.
- Gender: Helps identify segment-based churn patterns.
- Tenure: Represents customer relationship duration with the company.
- Usage Frequency: Reflects engagement level with the service.
- Contract Length: Indicates commitment level of the customer.
- Total Spend: Measures customer value and contribution to revenue.
- Target Variable (Churn): Binary outcome indicating whether a customer leaves or stays.

These features collectively help in building a robust predictive model.

Methodology:

The predictive modeling process was executed in IBM SPSS Modeler using the following steps:

1. Data Import:

The dataset was imported using a .var file through a source node to ensure proper structure and compatibility.

The screenshot shows the 'Data Source' dialog box in IBM SPSS Modeler, configured for importing a CSV file named 'customer_churn_dataset-training-master.csv'. The file path is 'C:\Users\Sri Harsha Uppala\Desktop\LPU_MBA_2025\Term4\INTM591\CA\Project\custome...'. The dialog includes a preview window showing the first few lines of the dataset, which include columns like CustomerID, Age, Gender, Tenure, Usage Frequency, Support Calls, Payment Delay, and Subscription. The 'Read field names from file' checkbox is checked. The 'Field delimiters' section shows 'Comma' and 'Newline' selected. The 'Quotes' section shows 'Pair and discard' selected for both single and double quotes. The 'OK' button is highlighted.

customer_churn_dataset-training-master.csv

Preview Refresh

C:\Users\Sri Harsha Uppala\Desktop\LPU_MBA_2025\Term4\INTM591\CA\Project\custome...

File Data Filter Types Annotations

File: C:\Users\Sri Harsha Uppala\Desktop\LPU_MBA_2025\Term4\INTM591\CA\Project\...

CustomerID, Age, Gender, Tenure, Usage Frequency, Support Calls, Payment Delay, Subs...

2, 30, Female, 39, 14, 5, 18, Standard, Annual, 932, 17, 1

3, 65, Female, 49, 1, 10, 8, Basic, Monthly, 557, 6, 1

4, 55, Female, 14, 4, 6, 18, Basic, Quarterly, 185, 3, 1

☒ Read field names from file ☐ Specify number of fields 1

Skip header characters: 0 EOL comment characters:

Strip lead and trail spaces: ☒ None ☐ Left ☐ Right ☐ Both

Invalid characters: ☒ Discard ☐ Replace with

Encoding: Stream default Decimal symbol: Stream default

☐ Line delimiter is newline character Lines to scan for column and type: 50

Field delimiters

☐ Space ☒ Comma ☐ Tab

☒ Newline ☐ Other

☐ Non-printing characters

☐ Allow multiple blank delimiters

☒ Automatically recognize dates and times

☐ Treat square brackets as lists

Quotes

Single quotes: Pair and discard

Double quotes: Pair and discard

OK Cancel Apply Reset

2. Data Preprocessing:

A Type Node was used to define variable roles and measurement levels, along with enabling the "Read Values" option.

The screenshot displays the IBM SPSS Modeler interface with the 'Type' node configuration window open. The 'Fields' tab is active, showing a list of fields on the left and their assigned roles and measurement levels on the right. The 'Targets' list contains 'Churn'. The 'Predictors (Inputs)' list includes 'Age', 'Gender', 'Tenure', 'Usage Frequency', 'Contract Length', and 'Total Spend'. The 'Analysis Weight' is set to 'Weight'.

Field	Measurement	Values	Missing	Check	Role
CustomerID	Continuous	[2,449,999]	None	None	None
Age	Continuous	[18,65]	None	None	Input
Gender	Nominal	Female...	None	None	Input
Tenure	Continuous	[1,60]	None	None	Input
Usage Freq...	Continuous	[0,10]	None	None	Input
Support Calls	Continuous	[0,30]	None	None	None
Payment Del...	Continuous	[0,30]	None	None	None
Subscription...	Nominal	Basic,P...	None	None	None

3. Model Building:

A CHAID decision tree model was developed to identify relationships between input variables and churn behavior.

The screenshot displays the IBM SPSS Modeler interface showing the CHAID model building process. The 'Churn' node is selected, and the 'Model' tab is active. The 'Model' tab shows the decision tree structure and the 'Predictor Importance' chart.

Predictor Importance
Target: Churn

The chart shows the importance of various predictors for the target variable 'Churn'. The x-axis represents the importance score from 0.0 to 1.0, with 'Least Important' on the left and 'Most Important' on the right. The y-axis lists the predictors: Total Spend, Age, Contract Length, Gender, Usage Frequency, and Tenure. 'Total Spend' is the most important predictor, followed by 'Age' and 'Contract Length'.

Model Summary

- Total Spend <= 404 [Mode: 1] => 1
- Total Spend > 404 and Total Spend <= 524 [Mode: 1]
- Total Spend > 524 and Total Spend <= 728 [Mode: 0]
- Total Spend > 728 and Total Spend <= 795 [Mode: 0]
- Total Spend > 795 [Mode: 0]

4. Model Testing:

A separate test dataset was imported and scored using the generated model nugget.

The screenshot shows the IBM SPSS Modeler interface. The 'File' menu is open, displaying options like 'Preview', 'Refresh', 'Generate', 'View', and 'Annotations'. The 'Churn' model is selected, and the 'Predictor Importance' chart is visible, showing the relative importance of various predictors like Total Spend, Age, and Gender. The chart indicates that Total Spend is the most important predictor, followed by Age and Gender. The interface also shows a list of model rules and a 'View: Predictor Importance' dropdown.

5. Selection of High-Risk Customers:

A Select Node was applied to filter customers with churn probability greater than 0.65.

The screenshot shows a table of customer data with 14 fields and 60,061 records. The columns include CustomerID, Age, Gender, Tenure, Usage Frequency, Support Calls, Payment Delay, Subscription Type, Contract Length, Total Spend, Last Interaction, Churn, SR-Churn, and SRC-Churn. The table is sorted by Churn probability, with the highest values at the top. The interface also shows a 'Table' tab and a 'Generate' button.

CustomerID	Age	Gender	Tenure	Usage Frequency	Support Calls	Payment Delay	Subscription Type	Contract Length	Total Spend	Last Interaction	Churn	SR-Churn	SRC-Churn
1	22	Female	25	14	4	27	Basic	Monthly	590	9	1	1	1.000
2	41	Female	28	28	7	13	Standard	Monthly	594	20	0	1	1.000
3	35	Male	9	12	5	17	Premium	Quarterly	232	18	0	1	1.000
4	53	Female	58	24	9	2	Standard	Annual	533	18	0	1	1.000
5	30	Male	41	14	10	10	Premium	Monthly	500	29	0	1	0.751
6	47	Female	37	15	9	25	Basic	Quarterly	574	14	1	0	0.747
7	54	Female	36	11	0	15	Standard	Monthly	323	16	0	1	1.000
8	36	Male	20	5	10	8	Basic	Monthly	687	8	0	1	1.000
9	65	Male	8	4	2	23	Basic	Annual	995	10	0	1	1.000
10	46	Female	42	27	9	21	Standard	Annual	526	3	1	0	0.734
11	56	Male	13	23	5	14	Basic	Quarterly	187	1	0	1	1.000
12	31	Male	2	7	0	25	Premium	Quarterly	750	24	0	0	0.779
13	59	Male	21	17	2	14	Premium	Quarterly	663	15	0	1	1.000
14	35	Female	1	3	7	3	Basic	Monthly	677	25	1	1	1.000
15	29	Male	54	3	6	2	Basic	Monthly	636	22	0	1	1.000
16	45	Male	9	30	4	25	Basic	Annual	127	18	0	1	1.000
17	65	Female	40	2	1	6	Premium	Annual	396	21	0	1	1.000
18	62	Male	39	19	2	15	Premium	Quarterly	202	24	0	1	1.000
19	48	Male	28	7	1	21	Premium	Monthly	925	13	0	1	1.000
20	36	Female	58	4	0	1	Premium	Quarterly	463	26	0	1	0.710
21	55	Male	50	28	0	17	Standard	Quarterly	449	3	0	1	1.000
22	36	Female	54	20	4	1	Basic	Monthly	373	25	0	1	1.000
23	64	Male	59	7	5	9	Basic	Annual	460	12	0	1	1.000
24	65	Female	55	7	3	30	Premium	Annual	166	1	1	1	1.000
25	53	Female	58	18	8	4	Basic	Quarterly	615	4	0	1	1.000
26	41	Female	60	7	7	0	Premium	Annual	696	20	0	0	0.734
27	44	Female	44	7	8	16	Basic	Quarterly	792	27	1	0	0.818
28	29	Female	23	8	8	0	Basic	Annual	812	3	0	0	0.755
29	34	Male	26	3	0	2	Basic	Annual	156	15	0	1	1.000
30	24	Male	1	7	0	3	Standard	Annual	611	4	0	0	0.782
31	27	Female	41	8	1	7	Premium	Monthly	943	12	0	1	1.000
32	31	Female	31	6	2	29	Standard	Quarterly	329	6	1	1	1.000
33	59	Male	44	17	3	2	Basic	Quarterly	636	30	0	1	1.000
34	56	Female	27	15	8	24	Standard	Monthly	354	6	1	1	1.000

6. Final Output Generation:

A Filter Node and Table Node were used to create a refined list of high-risk customers for business action.

Table (10 fields, 60,061 records)

File

Edit

Generate

Table

Annotations

	CustomerID	Age	Gender	Tenure	Usage Frequency	Subscription Type	Contract Length	Total Spend	Status	Probability	
1	1	22	Female	25	14	Basic	Monthly	595	1	1.000	
2	2	41	Female	28	20	Standard	Monthly	594	1	1.000	
3	3	4	35	Male	9	12	Premium	Quarterly	232	1	1.000
4	4	5	53	Female	58	24	Standard	Annual	933	1	1.000
5	5	6	30	Male	41	14	Premium	Monthly	500	1	0.781
6	6	7	47	Female	37	15	Basic	Quarterly	574	0	0.747
7	7	8	54	Female	34	11	Standard	Monthly	323	1	1.000
8	8	9	34	Male	20	3	Basic	Monthly	687	1	1.000
9	9	10	65	Male	8	8	Basic	Annual	995	1	1.000
10	10	11	46	Female	42	27	Standard	Annual	524	0	0.734
11	11	12	54	Male	13	23	Basic	Quarterly	187	1	1.000
12	12	13	31	Male	2	7	Premium	Quarterly	758	0	0.779
13	13	15	59	Male	21	17	Premium	Quarterly	663	1	1.000
14	14	16	15	Female	1	3	Basic	Monthly	677	1	1.000
15	15	17	29	Male	54	3	Basic	Monthly	636	1	1.000
16	16	18	45	Male	9	30	Basic	Annual	127	1	1.000
17	17	19	45	Female	40	2	Premium	Annual	396	1	1.000
18	18	20	62	Male	39	19	Premium	Quarterly	202	1	1.000
19	19	21	49	Male	28	7	Premium	Monthly	925	1	1.000
20	20	22	16	Female	88	4	Premium	Quarterly	463	1	0.714
21	21	23	59	Male	50	20	Standard	Quarterly	449	1	1.000
22	22	24	16	Female	54	20	Basic	Monthly	273	1	1.000
23	23	25	44	Male	59	7	Basic	Annual	460	1	1.000
24	24	26	63	Female	58	7	Premium	Annual	166	1	1.000
25	25	27	53	Female	50	19	Basic	Quarterly	615	1	1.000
26	26	28	41	Female	60	7	Premium	Annual	696	0	0.734
27	27	30	44	Female	44	7	Basic	Quarterly	792	0	0.818
28	28	31	28	Female	23	9	Basic	Annual	512	0	0.755
29	29	32	34	Male	26	3	Basic	Annual	156	1	1.000
30	30	33	24	Male	1	7	Standard	Annual	611	0	0.782
31	31	34	27	Female	41	9	Premium	Monthly	943	1	1.000
32	32	35	31	Female	31	6	Standard	Quarterly	329	1	1.000
33	33	37	59	Male	44	17	Basic	Quarterly	636	1	1.000
34	34	38	16	Female	27	15	Standard	Monthly	354	1	1.000
35	35	39	27	Female	44	21	Premium	Annual	515	1	0.751
36	36	40	42	Male	51	10	Basic	Monthly	520	1	1.000
37	37	41	29	Female	28	23	Basic	Quarterly	1000	0	0.785
38	38	42	16	Male	45	24	Premium	Quarterly	405	1	0.710
39	39	43	61	Male	19	2	Premium	Monthly	879	1	1.000
40	40	44	17	Male	2	20	Premium	Monthly	771	1	1.000
41	41	45	56	Male	9	16	Premium	Monthly	600	1	1.000
42	42	46	42	Female	14	14	Basic	Monthly	825	1	1.000
43	43	47	15	Female	24	21	Basic	Monthly	231	1	1.000

Streams

Outputs

Models

Stream1

StreamChannelModel

CRISP-DM

Classes

Business Understanding

Data Understanding

Data Preparation

Modeling

Evaluation

Deployment

CRISP-DM

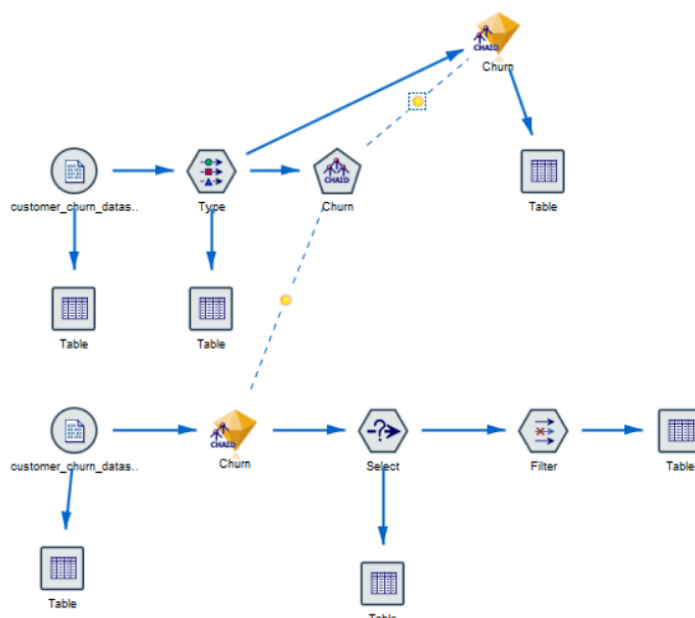
Classes

OK

153MB / 253MB

Model Insights:

The CHAID (Chi-square Automatic Interaction Detection) algorithm is effective for segmentation and classification problems. It splits data based on statistically significant relationships between variables, making it highly interpretable for business users.



The chosen threshold of 0.65 confidence ensures a balance between accuracy and actionability. A higher threshold reduces false positives, allowing the business to focus only on customers with a strong likelihood of churn. This improves the efficiency of retention campaigns and optimizes resource allocation.

Conclusion:

The project successfully identifies customers with a high probability of churn using predictive analytics. The final output table provides a targeted list that can be directly used by the marketing team.

Recommendations:

- Implement personalized retention campaigns for high-risk customers.
- Offer discounts or loyalty benefits to customers with low tenure and usage.
- Monitor customer engagement regularly to detect early churn signals.
- Use predictive models periodically to update churn risk profiles.

By acting on these insights, organizations can significantly reduce churn rates and enhance customer retention.